

### 3.2.5 Resource security

This optional section of our specification focuses on the large-scale exploitation of unevenly distributed natural resources, which is one of the defining features of the present era. Increasing demand for water, energy and minerals and their critical role in human affairs leads to massive local and regional transfers of water and massive global transfers of energy and minerals.

In this section students contemplate the fundamental relationships between the physical environment and human activities and wants and the relationships between people in their local, national and international communities involving themes of sustainability and conflict. They engage with these themes in relation to energy, water and minerals but may concentrate on one or other in their case studies.

Study of this section offers the opportunity to exercise and develop observation skills, measurement and geospatial mapping skills, together with data manipulation and statistical skills, including those associated with and arising from fieldwork.

### 3.2.5.1 Resource development

Concept of a resource. Resource classifications to include stock and flow resources. Stock resource evaluation: measured reserves, indicated reserves, inferred resources, possible resources. Natural resource development over time: exploration, exploitation, development. Concept of the resource frontier. Concept of resource peak.

Sustainable resource development. Environmental Impact Assessment (EIA) in relation to resource development projects.

### 3.2.5.2 Natural resource issues

Global patterns of production, consumption and trade/movements of energy and ore minerals. Global patterns of water availability and demand.

The geopolitics of energy, ore mineral and water resource distributions, trade and management.

### 3.2.5.3 Water security

Sources of water; components of demand, water stress.

Relationship of water supply (volume and quality) to key aspects of physical geography – climate, geology and drainage.

Strategies to increase water supply to include catchment, diversion, storage and water transfers and desalination.

Environmental impacts of a major water supply scheme incorporating a major dam and/or barrage and associated distribution networks.

Strategies to manage water consumption (including reducing demand).

Sustainability issues associated with water management: virtual water trade, conservation, recycling, 'greywater' and groundwater management.

Water conflicts at a variety of scales – local, national, international.

### 3.2.5.4 Energy security

Sources of energy, both primary and secondary. Components of demand and energy mixes in contrasting settings.

Relationship of energy supply (volume and quality) to key aspects of physical geography – climate, geology and drainage.

Energy supplies in a globalising world: competing national interests and the role of transnational corporations in energy production, processing and distribution.

Environmental impacts of a major energy resource development such as an oil, coal or gas field and associated distribution networks.

Strategies to increase energy supply (oil and gas exploration, nuclear power and development of renewable sources).

Strategies to manage energy consumption (including reducing demand).

Sustainability issues associated with energy production, trade and consumption: acid rain, the enhanced greenhouse effect, nuclear waste and energy conservation.

### 3.2.5.5 Mineral security

With reference to iron ore **or** a specified globally traded non-ferrous metal ore eg copper, tin, manganese.

Sources of the specified ore. Distribution of reserves/resources. End uses of the ore. Components of demand for ore. Role of specified ore in global commerce and industry.

Key aspects of physical geography associated with ore occurrence and working: geological conditions and location.

Environmental impacts of a major mineral resource extraction scheme and associated distribution networks.

Sustainability issues associated with ore extraction, trade and processing.

### 3.2.5.6 Resource futures

Alternative energy, water and mineral ore futures and their relationship with a range of technological, economic, environmental and political developments.

### 3.2.5.7 Case studies

**Case study** of **either** water **or** energy **or** mineral ore resource issues in a global or specified regional setting to illustrate and analyse theme(s) set out above, their implications for the setting including the relationship between resource security and human welfare and attempts to manage the resource.

**Case study** of a specified place to illustrate and analyse how aspects of its physical environment affects the availability and cost of water **or** energy **or** mineral ore and the way in which water or energy or mineral ore is used.